

Emerging Careers and Their Implications

In July 2024, China's Ministry of Human Resources and Social Security introduced 19 new careers, including roles like biotechnologists and generative AI specialists. These emerging fields offer significant opportunities for graduates, shaping the future workforce beyond just technical students. Choosing a career that aligns with personal growth becomes crucial. Our mathematical model aims to analyze **career prospects** and long term development trend, providing guide to graduates in selecting paths suited to their backgrounds and goals.

Our first model measures the prospects of new careers. It is based on the **OECD Framework for Measuring and Assessing Job Quality**, which provides specific indicators for career prospects. After adjusting several indicators to align with the Chinese labor market, we identified a total of **12 indicators** (4 primary and 8 secondary) encompassing various aspects of job prospects. Despite the fact that emerging careers have limited data, we collected a database of **12 data points**. Although there are significant differences in data magnitude, our normalization method effectively addresses this challenge. Our predictions for smaller-scale careers are particularly precise due to the comprehensive data collection. We applied our model to five emerging careers (Intelligent Connected Vehicle Tester, Intelligent Connected Vehicle Installation And Maintenance Technician, Cloud Network Intelligent Maintenance Technician, User Growth Operations Specialist, and Live Streamer) (using the **TOPSIS** calculation method. The results indicate that the **Cloud Network Intelligent Maintenance Technician** has the highest overall career prospect, followed by the **Live Streamer**.

However, a simple understanding of the career prospect is not enough to determine how a career will go. Therefore, we modify our model, deleting variables that are related to immediate career prospects and adding some that relate to the **long term development trend** (such as **fungibility, government policy support, and talent development**). We expand our search of careers to all 19 emerging careers, which encompasses a more comprehensive database. We switch to **AHP method** to calculate the weights due to the abundance of indicators. After **TOPSIS** calculation, the top five careers with highest development trend scores are **Non-Ferrous Metal Spot Trader, User Growth Operations Specialist, Intelligent Manufacturing System Maintenance Technician, Industrial Internet Maintenance Technician, and Cybersecurity Grading and Assessment Specialist**.

Finally, we shift our focus to the **individual graduate** seeking a career in emerging fields. We developed a **personalized survey** that gathers key information about each graduate's **background and interests**. Using this data, along with our model's previous findings, we recommend careers with the **top 3 development trends**. This approach ensures that our model aligns not only with the graduate's preferences but also with market trends, providing a tailored fit to both their goals and industry demands.

Keywords: Career Prospect; Emerging Careers, Career Selection

Contents

1 Introduction	4	Trends of Emerging Professions	13
1.1 Problem Background	4	4.1 Indicators	14
1.2 Problem Statement and Analysis . . .	4	4.1.1 Industry Potential	14
2 Assumptions	5	4.1.2 Industrial Transformation . . .	14
3 Model For Career Prospect	5	4.2 Database	14
3.1 Factors explanation	5	4.3 Normalization	15
3.2 Career selection	6	4.4 Indicator Weights and AHP	15
3.3 Indicators	6	4.4.1 Methodology	15
3.3.1 Market Supply-Demand Balance	6	4.4.2 AHP Process and Results . . .	16
3.3.2 Income Level	7	4.5 TOPSIS	17
3.3.3 Developing Potential	8	4.6 Results and Analysis	17
3.3.4 Working conditions and stability	8	4.6.1 Career Trend Result	17
3.4 Database	9	4.6.2 Analysis	17
3.5 Re-expressing Data and Normalization	10	4.7 Sensitivity Analysis	19
3.6 Coefficient of Variation (CV) Method	11	5 Strength and Weakness	19
3.7 TOPSIS	12	6 Providing Career Advice for Graduates	19
3.8 Results	13	6.1 Individual Survey and Example	19
3.9 Sensitivity Analysis	13	6.2 Matching the Students With The Careers	20
4 Model for Predicting the Development		6.3 Strengths and Weaknesses	20
		7 Science Article	20
		8 Conclusion	21
		A Appendix	25

1 Introduction

1.1 Problem Background

The topic of career choice has been a subject of considerable discourse for a long time in China. However, it has gained even greater urgency as the unemployment rate for urban youth (ie. 16 to 24 years old) rose to 18.8% in August 2024, the highest figure for the year. (See Figure 1.1) This trend follows a previous peak of 21.3% in June 2023, after which the government revised its methodology, excluding students from the calculation[20]. Despite these changes, the youth unemployment rate has remained persistently high. In such a challenging job market, selecting the right career path becomes essential.

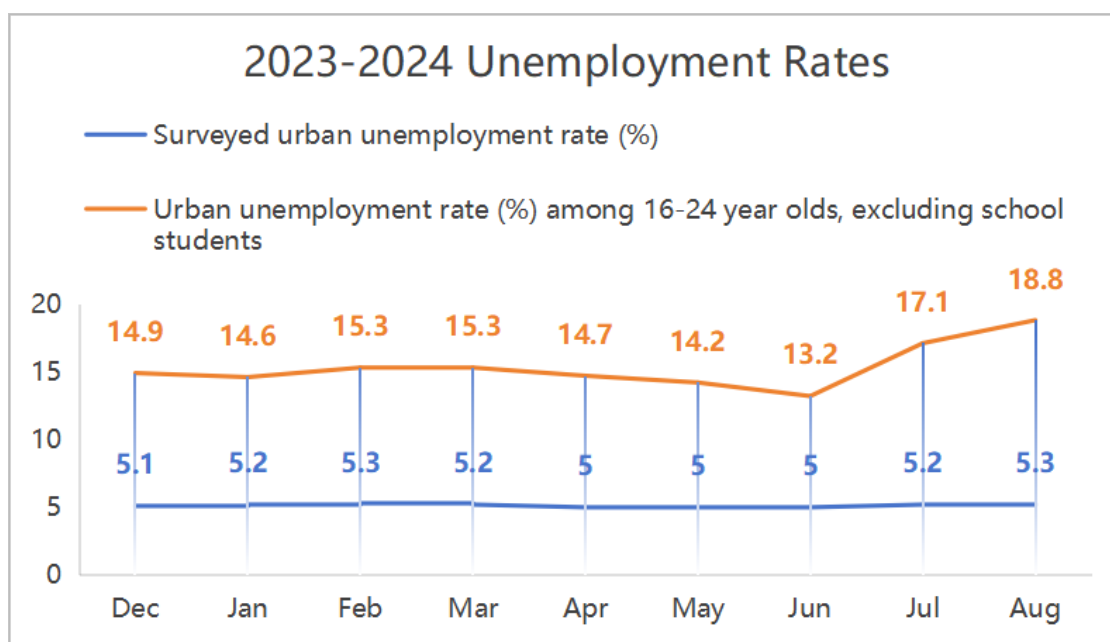


Figure 1.1: Urban Unemployment Rate versus Unemployment Rate of 16-24-Year-Olds

In response to the urgent employment situation, in July 2024, China's Ministry of Human Resources and Social Security, in cooperation with the State Administration for Market Regulation and the National Bureau of Statistics, officially introduced 19 new professions. These roles, including Bioengineering Technicians and Generative AI System Application Specialists, cover a broad range of sectors such as bioengineering, cybersecurity, artificial intelligence, intelligent manufacturing, and renewable energy[21]. These professions align with the economic trend, but they also present uncertainties regarding prospects and future potential.

1.2 Problem Statement and Analysis

With increased competition and fewer immediate job openings in established fields, it's essential for graduates to consider not just immediate employment but also the long-term prospects and potential growth of their chosen careers.

Our thought process centers on addressing this multifaceted challenge by developing a comprehensive mathematical model. We recognize that making informed career decisions requires analyzing multiple dimensions beyond surface-level job availability. Factors such as market demand, income levels, working conditions, development potential, and future industry trends play crucial roles in shaping a profession's viability.

To determine the problem, we began by identifying the gaps in current career guidance, particularly concerning emerging professions that lack historical data. We considered how we could fill these gaps by quantifying various factors that influence career prospects. Our aim is to create a model that not only assesses the current state of emerging professions but also predicts their future development over the next decade. Therefore, in order to model the long-term trend of the new careers, we modify our indicators to again quantify the perceived growth of different careers.

Furthermore, we understand that each graduate's situation is unique, influenced by their academic back-

ground, geographic location, and personal interests. Therefore, our model incorporates personalized elements to align individual profiles with the assessed career prospects, providing tailored advice that resonates with each graduate's aspirations and circumstances. We interpret the problem as the following fundamental questions:

Questions.

- (1) What are the **key factors** influencing the prospects of emerging professions, and how can we quantitatively assess these factors to provide a comprehensive career rating for the 5 selected professions?
- (2) What are the predicted development trends of the 19 newly emerging professions over the next decade
- (3) Which professions are likely to gain popularity and maintain relevance in the future, and how can our predictive model analyze long-term trends to identify these professions?
- (4) How can we integrate individual graduate backgrounds—such as major, region, and career interests—with the assessed career prospects to offer personalized career advice?

2 Assumptions

Assumption 1: Total market economy in a certain region or country has a significant impact on employment. In order to make the stimulation more precise, we assume there is no sudden increase or decrease in the economy because of devastating issues, such as pandemics, war, etc..

Assumption 2: Since all of the fields are developing at the first stage, monopoly and other unhealthy market structure might be developed as the time elapsed, which causes the significant difference between the actual and simulated data. Thus, it is necessary to assume all of the markets are competitive all the time.

Assumption 3: Since the market composition around the world can be various in different countries, to make the data collection easier and ignore the global political impacts on fields, we assume all of the jobs discussed are developing in the Chinese mainland.

Assumption 4: To minimize the impact except the market itself, we assume the governments' incentive policies remain stable all the time.

Assumption 5: To stimulate the results precisely by using the model, we assume the time period for *Model For Career Prospect* is 2 years, while the time period for *Model for Predicting the Development Trends of Emerging Professions* is 10 years, over the long term.

Assumption 6: For *Model for Providing Career Advice for Graduates*, we ignore some minor impacts, including conditions and environments around the business, working time, and personal bias to working conditions.

Assumption 7: To adjust the results due to the changing personal interest to each job, the model: *Model for Providing Career Advice for Graduates* can be used several times whenever the graduates change their interest.

Assumption 8: In *Model for Providing Career Advice for Graduates*, we assume all of the job-seekers have graduated from the colleges, which have no difference in their education.

3 Model For Career Prospect

3.1 Factors explanation

To evaluate the final career prospects, we have chosen the Supply and Demand of a career's market, the Income level of a career, the development potential of a market, and the Working conditions of a market as our indicators. The supply and demand and income level decide if there will be skilled workers entering the market of a career, and the working conditions decide if people will accept the job after entering the career market. The developing potential of a career decides whether the career is going to have a growing trend in the future or not.

3.2 Career selection

To give a general view of career prospects, we choose 5 careers from the 19 new professions that are considered to be **representative** in the sense that they represent the emerging trend of the career market: automotive technology, cloud computing, digital content creation, smart manufacturing, and intelligent transportation.

The careers we choose and basic descriptions are illustrated in Table 4.1.

Career	Career Description
Intelligent Connected Vehicle Tester	Conducts functional verification and testing of intelligent connected vehicles and related components.
Cloud Network Intelligent Maintenance Technician	Engages in the operation and maintenance of cloud network service systems using cloud computing and intelligent network technologies.
Live Streamer	Provides online performances and audio-visual information services through live streaming and real-time interaction.
Intelligent Manufacturing System Maintenance Technician	Performs data collection, fault diagnosis, and preventive maintenance for intelligent manufacturing systems.
Intelligent Connected Vehicle Installation and Maintenance Technician	Assembles, adjusts, tests, and maintains intelligent connected vehicles and roadside equipment.

Table 3.1: Career Introductions

3.3 Indicators

Figure 4.1 provides a comprehensive table of indicators used in the paper.

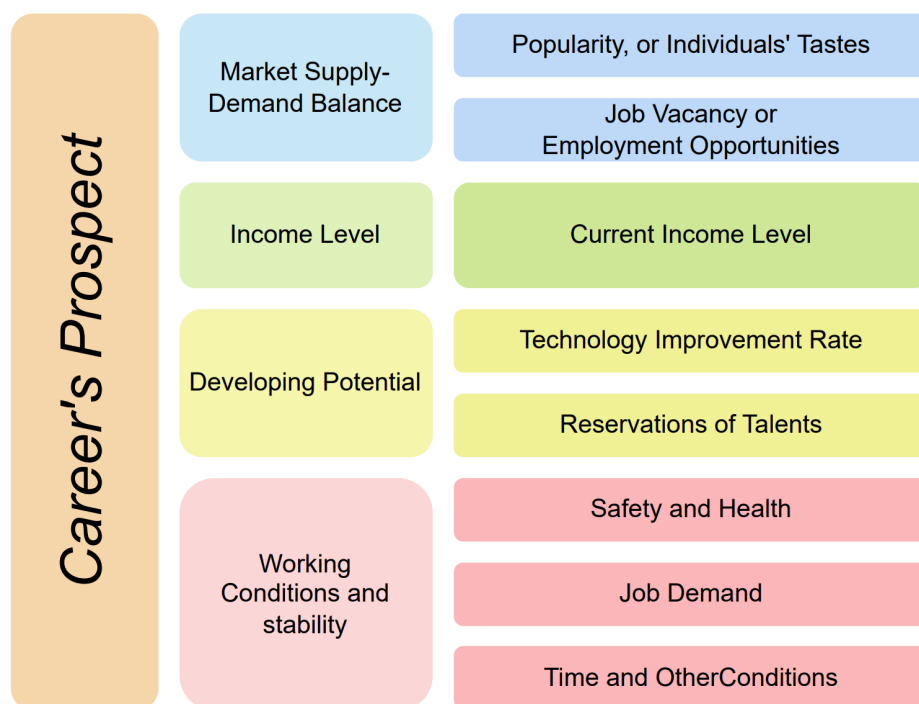


Figure 3.1: Primary and Secondary Indicators

3.3.1 Market Supply-Demand Balance

We deem the market supply-demand ratio to be a critical factor in determining career prospects after research on Economics. In labor economics, the market supply-demand balance for jobs depends on the interaction between labor supply (workers available for a particular job) and labor demand (employers looking

for workers to fill vacancies). In-demand jobs are more likely to lead to higher career prospects due to the increased need for specialized skills. For instance, studies have shown a transformation in labor demand in China, more specifically an increase in demand for computer and healthcare industries and a decline in real estate-related careers[1]. This shift in demand is predicted to change the preferences when companies assess potential employees, to the extent of influencing choices of majors in universities. Below are some secondary indicators that we deem essential for determining the market supply-demand balance.

a. Popularity, or Individuals' Tastes

A broader review of employability skills demonstrates that popular career paths tend to align with the needs of the modern economy[4]. A way to understand this is by applying the demand curve from microeconomics. A demand curve, in typical economics, is an indicator of "how much of good consumers are willing to buy as the price per unit changes[2]." One of the important *Determinants of Demand* states is change in consumer's tastes[6]. This indicates that an increase in consumers' tastes will shift the demand curve to the right, as shown in Figure 10.1 [2].

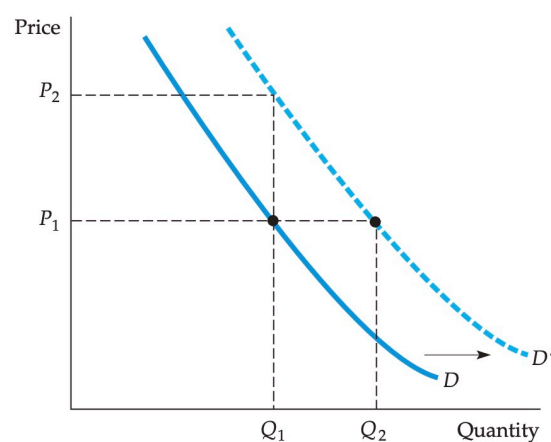


Figure 3.2: Shift in the Demand Curve as a Result of Consumers' Tastes

The same logic can be applied to the demand for jobs in certain career paths. When a particular career becomes more aligned with the interests, needs, and tastes of job seekers (eg. due to increased media attention or perceived job security), the "demand" for these roles also increases. This will significantly influence the prospects of a job in multiple ways, such as providing more available positions and introducing more technological improvements.

b. Job Vacancy or Employment Opportunities

Job vacancies represent the unfilled positions within a market, highlighting areas where labor demand exceeds supply. In labor economics, the interaction between labor supply (the number of job seekers) and labor demand (the number of available positions) determines the equilibrium in the job market. A high number of vacancies for job seekers indicates a tight labor market, often leading to increased wages and better employment conditions as employers compete for jobs. Conversely, a surplus of job seekers over available positions signals a slack labor market, potentially leading to unemployment or underemployment. Incorporating job vacancy data into our mathematical model allows us to capture these nuances. By analyzing the vacancy gap, we can assess not only the current state of the labor market but also project future trends in employment, wages, and career growth opportunities. This approach aligns with the principles outlined by Blanchard and Diamond (1989), who emphasized that "the relationship between unemployment and vacancies is a key indicator of labor market efficiency and dynamics[9]."

3.3.2 Income Level

Current Income Level

Income level is a fundamental determinant of job attractiveness. Higher-paying professions naturally draw more interest from job seekers due to the immediate financial benefits. According to the Human Capital Theory proposed by Becker (1964), individuals are motivated to invest in education and skill development when they

anticipate higher returns in the form of increased earnings[14]. Therefore, a higher income level serves as a major incentive for individuals to enter and remain in a profession, which enhances the talent pool and overall competitiveness within that field. Moreover, income level is also correlated with opportunities for technological growth and advancements. The current income level reflects the investments and financial support in a field, so a higher income level usually comes with better resources for development and better training.

3.3.3 Developing Potential

a. Technology Improvement Rate

While technology has been intensively improved due to online and digital development, more jobs and fields become unsaturated. Therefore, the progress of upgrading technology leads to more available jobs to employees, as a benefit. According to a previous study, new technology can lead to the renovation of capital goods, eliminating old jobs, and creating new jobs. The determinant is the cost of this process, including research, experiments, and transportation. In particular, low cost can bring more benefits to the labor market, which decreases unemployment[15]. Furthermore, while digital technology develops, low-skilled workers' demand is decreasing because machines can effectively substitute low-skilled workers[16].

b. Reservations of Talents

Talents too can affect any market or field. A sufficient supply of talent can provide sustainable support to related goods or services. Consequently, this may create more opportunities for more employees. Even though the previous research is based on a relatively small amount of data in a limited range, it is plausible to illustrate that talent managing or training has positive correlations with corresponding services[17].

3.3.4 Working conditions and stability

a. Safety and Health

Safety is a topic that people consider before anything since we assume all individuals are rational. It is self-evident that safety is an indispensable factor to be considered. Thus, high-risk jobs always will be seriously considered if it is really worth the risk. Generally, despite the working environment, which will be discussed in a later passage, absenteeism can relatively reflect the hazardous working conditions which are positively related. One of the disadvantages of this research is unpredictable and private variables affecting absenteeism, including age, race, wages, etc.[5]. Also, mental health is considered, as many works may do mental damage to people.

b. Job Demand

Job demand is an important factor in measuring stability and working conditions as well. It fundamentally affects various aspects of employees' lives, such as their own well-being, productivity, and adaptability. It is a subjective standard based on everyone's personal dealings. Job demand contains factors of both physical and psychological working ability. Generally, job demand can be seen as a challenge as well, which has to be optimized instead of decreased, balancing resources and demand[7]. This factor correlates with health tightly.

c. Time and Other Conditions

I. Working Time and Working Intensity

1. Working Time

Being mentioned before, the difficulty of any job has to be weighed before choosing, as well as habits. Various people have numerous time schedules, which reveals they may have different suggested time periods of working. Thus, time is one of the working conditions that have to be weighted. Efficiency always decreases when time elapses, related to the total time in a given period, according to previous research[8]. The figure illustrates the principle of diminishing marginal returns, which shows the relationship between hours worked and productivity. Initially, productivity increases with more hours worked, but after a certain point, the rate of productivity gain slows down significantly, demonstrating diminishing returns. Beyond this point, additional working hours lead to minimal or even negative productivity gains due to factors like fatigue, decreased focus, and reduced efficiency.

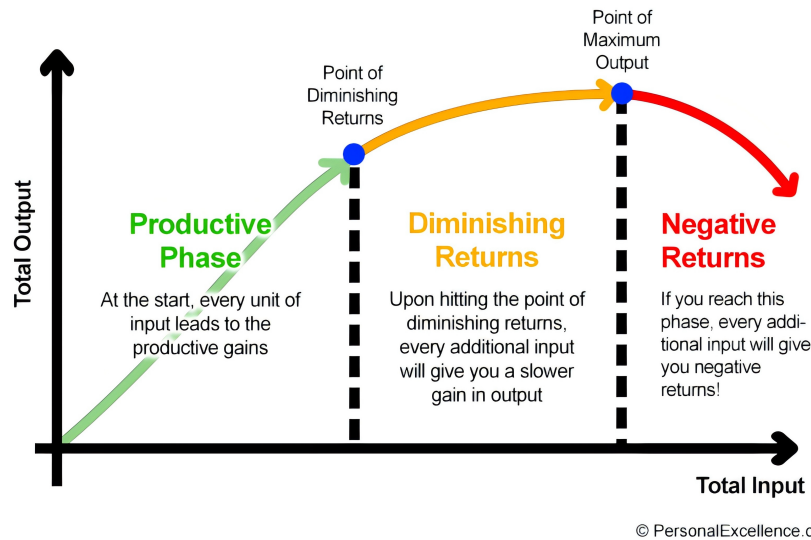


Figure 3.3: Principle of Diminishing Marginal Returns[10]

2. Intensity of Work

Nowadays, there is an online buzzword called '996', meaning working from 9 a.m. to 9 p.m. in six days out of a week. This reveals the high working intensity of employees. Related to the time factor, the intensity of work can be defined as the total working time in a unit period. The working intensity was found to be positively correlated with working pressure and status[11].

II. Working Period

Under the assumption that all employees are mostly active in the daytime, and inactive at night, there should be differences in efficiency between different periods of time. According to a previous study, working during the night may increase subjective sleepiness and reaction time. Specifically, no matter how long employees have rest before the night shift, there is a significant decrease in performance and clear-headedness[12]. Thus, it is necessary to consider the fact of tiredness during the night shift, while a certain amount of work has to be accomplished, which employees have to overcome this challenge.

III. Social Conditions

The impacts of society on jobs are quite obvious and essential. Specifically, the impacts of racism, harassment, and other unethical actions can heavily bias the decisions that employees make. Under pressure from both society and work, employees perform poorly and display worse results than the others. On the other hand, considering the positive impacts of society, better social relationships with others can lead to better performance, according to the previous study. Using deep analysis of multiplex networks of employees and the UMN model, employees with high centrality in a weighted UMN are more likely to perform well[13].

3.4 Database

Based on our indicators, we find retrospective data for each criterion. While some data are directly addressed online, others need further calculation and predictions.

There are two aspects, or secondary indicators for supply-demand balance: popularity and current job demand. **Popularity**, by definition, is the degree of public interest and awareness surrounding a profession. **Frequency of searches and discussions** is deemed to be effective in evaluating public interest. When people frequently search for or talk about a topic, it indicates curiosity and relevance to which the subject is capturing attention. In order to capture **immediate spikes of interest**, **sustained interest over time**, and **how often career-related content is viewed on a major platform**, we measure weekly search volume, monthly search volume, and Baidu views respectively.

Similarly, income level is divided into **low average salary** and **high average salary**. Developing Potential is divided into **impact of technological advancement**(growth rate of market size) and **talent cultiva-**

tion(industry talent supply index). They are used to represent different aspects of the same secondary indicator.

Selected Career	Supply-Demand Balance			
	Popularity			Current Job Demand
	Weekly Search Volume	Baidu entry view count	Related web pages (thousands)	Job Vacancy(10 thousand)
Intelligent Connected Vehicle Tester	98	1249	1630	1.87
Intelligent Connected Vehicle Installation and Maintenance Technician	35	461	135	1.83
Cloud Network Intelligent Maintenance Technician	28	988	869	150
User Growth Operations Specialist	147	1755	455	150
Live Streamer	434	727820	10300	2184.18

Selected Career	Working Conditions and Stability			
	Security	Condition	Standard Working Time	
		Efficiency	Daily Work Time(hours)	Avg Work Time
Intelligent Connected Vehicle Tester	1	1	5.71	40hours/week
Intelligent Connected Vehicle Installation and Maintenance Technician	1	1	5.71	40hours/week
Cloud Network Intelligent Maintenance Technician	1	1.253	5.71	40hours/week
User Growth Operations Specialist	1	1.253	5.71	40hours/week
Live Streamer	1	1.253	4.74	144hours/week

Selected Career	Income Level		Developing Potential	
	Current Salary		Impact of Technological Advancement	Talent Cultivation
	Low Avg Salary (thousand)	Average high salary(thousands)	Average Growth Rate of Market Size	Industry Talent Supply Index
Intelligent Connected Vehicle Tester	9	18	0.51	60
Intelligent Connected Vehicle Installation and Maintenance Technician	8	30	0.51	60
Cloud Network Intelligent Maintenance Technician	10	80	0.2	98.4
User Growth Operations Specialist	8	8	0.11	98.4
Live Streamer	3	5	0.55	98.4

Figure 3.4: Data of the Career Prospect Used in the Paper

3.5 Re-expressing Data and Normalization

Next, we re-express the data for income level by taking a **harmonic mean** of low and high average income levels. The formula is expressed as follows:

$$I_{\text{income}} = \frac{2 \cdot I_{\text{low}} \cdot I_{\text{high}}}{I_{\text{low}} + I_{\text{high}}} \quad (3.1)$$

where I_{low} is the low average income level, I_{high} is the high average income level, and I_{income} is the harmonic mean of income.

In income distributions, which are often right-skewed (a small proportion of people earn much higher incomes than the majority), the harmonic average accounts for the **disproportionate effect of lower salaries**. By emphasizing the lower range of income, the harmonic average provides a more balanced measure of income level than the arithmetic mean. Next, we apply different normalization methods to determine the value of secondary indicators.

The indicators are normalized using two methods: **Ratio Transformation Method** and **Inverse Normalization**. The Ratio Transformation Method is applied to criteria where higher values indicate better performance, commonly referred to as benefit criteria. These include **Weekly Searches**, **Baidu Views**, **Related Webpages**, **Talent Gap**, and **Efficiency**. In contrast, **Inverse Normalization** is applied to **Daily Working Hours**, which is a cost criterion where lower values are preferred as they reflect better working conditions. This method inversely scales the data to penalize higher values.

For the **Ratio Transformation Method** (applied to Weekly Searches, Baidu Views, Related Webpages, Talent Gap, and Efficiency):

$$x_1^{\text{normalized}} = \frac{x_1}{\max(x_1)} \quad (3.2)$$

where x_1 represents the original value of the indicator, and $\max(x_1)$ is the maximum value in the column.

For **Inverse Normalization** (applied to Daily Working Hours):

$$x_2^{\text{normalized}} = \frac{\min(x_2)}{x_2} \quad (3.3)$$

where x_2 represents the original value of the indicator, and $\min(x_2)$ is the minimum value in the column.

3.6 Coefficient of Variation (CV) Method

We apply **Coefficient of Variation (CV) Method** to calculate the weight of each respective category.

CV is a data-driven technique used to determine the relative weights of criteria in multi-criteria decision-making. It assesses the importance of each criterion based on the degree of variability in its data values. A higher coefficient of variation indicates that the criterion has greater dispersion or variability, suggesting stronger discriminatory power among alternatives and thus higher importance. The method follows these steps:

1. **Compute the Mean (μ_j) and Standard Deviation (σ_j)** for each criterion j :

$$\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \mu_j)^2} \quad (3.4)$$

where x_{ij} is value of the j -th criterion for the i -th alternative, and n is the number of alternatives.

2. **Calculate the Coefficient of Variation (CV_j)** for each criterion:

$$CV_j = \frac{\sigma_j}{\mu_j} \quad (3.5)$$

where σ_j is the standard deviation of the j -th criterion, and μ_j is the mean of the j -th criterion.

3. **Determine the Weight (w_j)** for each criterion by normalizing the CV_j values:

$$w_j = \frac{CV_j}{\sum_{j=1}^m CV_j} \quad (3.6)$$

where m is the total number of criteria.

The value for each secondary indicator is calculated as the weighted sum of all respective categories. This is achieved by multiplying the value of each category by its corresponding weight and summing them together. For example, the final quantitative result of popularity is calculated using the following formula:

$$S_P = w_1 \times S_w + w_2 \times S_m + w_3 \times V_b \quad (3.7)$$

where w_1 , w_2 , and w_3 are the weights determined by the Coefficient of Variation (CV) Method.

The respective weights of different indicators are illustrated in Figure 3.5.

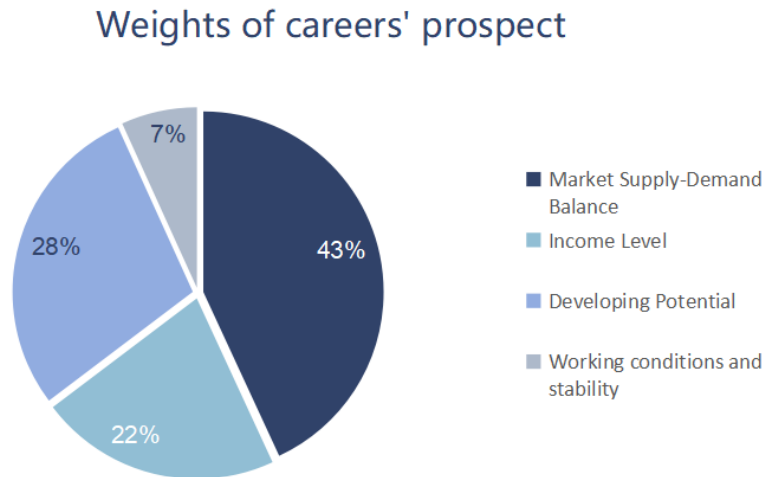


Figure 3.5: Weights of Careers' Prospect

3.7 TOPSIS

We apply **TOPSIS** to calculate the final results of Career Prospect.

TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) is a multi-criteria decision-making method used to rank alternatives based on their proximity to an ideal solution. It identifies the best and worst possible solutions, known as the ideal and negative-ideal solutions, based on specific criteria. The alternatives are then evaluated by calculating their distance from both the ideal and negative-ideal solutions. The alternative closest to the ideal solution is ranked highest. This model is valuable in assessing career trends because it allows us to measure how good a career's prospects are by comparing its characteristics—such as market demand balance, industry potential, and industry transformation—with the ideal scenario. A TOPSIS demonstration is indicated in the Figure3.6.

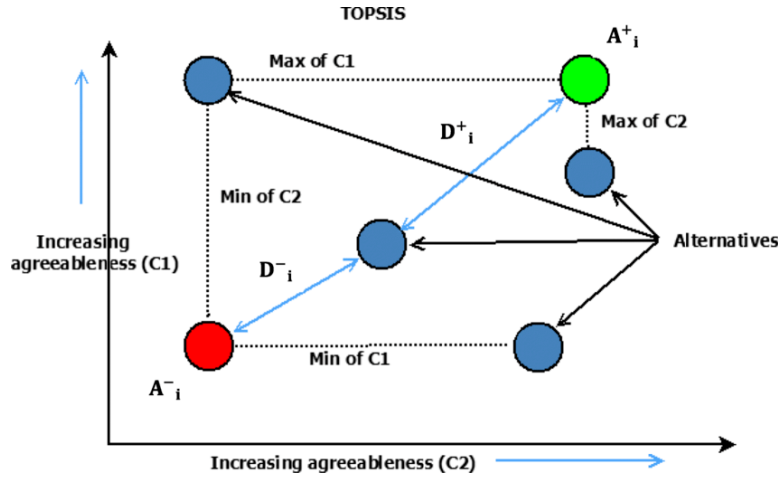


Figure 3.6: Illustration of TOPSIS[33]

First, the decision matrix is normalized, where each value x_{ij} (the value of the i -th alternative for the j -th criterion) is transformed into a normalized value r_{ij} :

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}$$

Next, a weighted normalized decision matrix is calculated by multiplying each r_{ij} with the weight w_j of the corresponding criterion:

$$v_{ij} = w_j \cdot r_{ij}$$

Then, the ideal and negative ideal solutions are determined. The **ideal solution** A^+ is composed of the best possible values for each criterion, and the **negative-ideal solution** A^- consists of the worst possible values:

$$v_j^+ = \max(v_{ij}) \quad \text{for benefit criteria, and} \quad v_j^- = \min(v_{ij}) \quad \text{for cost criteria}$$

Finally, the distance of each alternative from the ideal solution D_i^+ and the negative-ideal solution D_i^- is computed:

$$D_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2} \quad \text{and} \quad D_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2}$$

The relative closeness of each alternative to the ideal solution is then calculated as:

$$C_i^+ = \frac{D_i^-}{D_i^+ + D_i^-}$$

The alternative with the highest C_i^+ value is considered the best option, as it is the closest to the ideal solution and farthest from the negative-ideal solution.

3.8 Results

After calculation, The final result is shown in Figure 3.7.

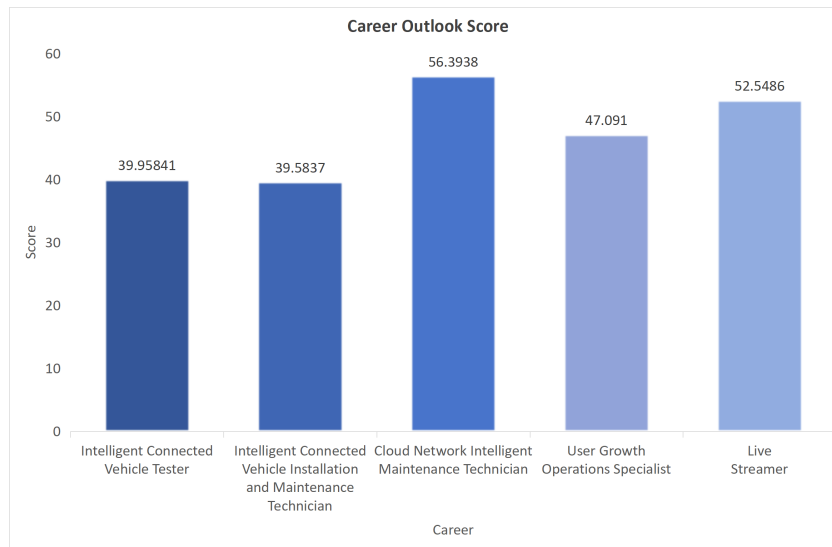


Figure 3.7: Final Score of Career Prospect

3.9 Sensitivity Analysis

To examine whether our results are sensitive to the weights of variables, we adjusted the original values of these weights to observe how the outcomes would change. The table below illustrates the sensitivity of our results with respect to changes in weights ($\pm 10\%$). Positive sensitivity values indicate that a change in the indicators' weight leads to an increase in the scores of jobs, while negative values signify that the output decreases when the weight changes.

Indicator	+10% Change	-10% Change
Weekly search volume	4.28	-3.72
Baidu entry views	1.12	-0.57
Related webpages	0.09	-0.03
Talent gap	1.33	-1.26
Average salary	1.00	-0.88
Market size growth rate	-3.43	3.05
Talent supply index	-5.20	4.90
Daily working hours	0.07	-0.02

Table 3.2: Sensitivity Analysis Results

The results of the sensitivity analysis indicate that the model is largely robust to variations in most indicators, with a few exceptions such as "Talent supply index" and "Market size growth rate." A high level of sensitivity for these indicators may be reasonable, as these indicators likely represent the core drivers of the system being modeled. However, our model also has a high sensitivity to "Weekly search volume", this is unexpected for us, and we should be aware of this when using this model.

4 Model for Predicting the Development Trends of Emerging Professions

After analyzing the career prospects of the five emerging professions, we are eager to explore the long-term development trends of these careers.

Accordingly, we have made some adjustments to the indicators in our model. Specifically, we have added **fungibility, government policy support, and talent development** as new factors, while **removing current income levels, working conditions, and job security** from consideration.

4.1 Indicators

4.1.1 Industry Potential

The industrial transformation of a market often affects its serving careers. For example, during the Industrial Revolution, the change from weaving in hands to weaving with machines caused a lot of people to lose their jobs[24]. In economics, structural unemployment often happens[5]. The policies often affect careers too, the government's and firm's decisions on EVs in China is an example[25]. Considering these factors, we choose to make the industrial transformation an indicator of career development trend.

Fungibility

Fungibility be basically explained as the plausible percentage of using technology instead of labor, which brings unemployment to fields. Thus, fungibility should be considered as a negative factor in the simulating model.

Due to technology's progress, more machines and artificial intelligence can substitute for low-skilled workers. An increasing number of capital goods, goods used for future production and transportation, causes the demand for labor to decrease intensively. According to a previous study, it is proved that technological improvement somehow affects and lessens the employment rate of low-skilled demand, which leads to a higher unemployment rate. Precisely, in the long term, the total cost of the capital goods is less than the salary distributed to employees under the law of minimum wage. It is more economical for businesses to purchase machines and digital technology to substitute low-skilled labors.[23].

4.1.2 Industrial Transformation

a. GDP's influence

The GDP trend of an industry reflects the current demand of a market. The GDP is the total value of products produced in that industry. As we just mentioned, if the demand increases in a market, the output will increase in that market, leading to a higher industry GDP; on the other hand, if the market's demand decreases, the output will drop, decreasing the industry's GDP[5]. Therefore, the GDP trend reflects people's willingness to buy in this market, making it one of our indicators for social need; so it is worth simulating the future trend of the GDP, to speculate the trend of employment growth. In order to show GDP's influence more directly, we focus on each market's market value. The higher the value of a market, the more likely firms are likely to enter[2]. As more firms enter a market, the careers in that market will increase in amount or increase in wage[5]. If the trend of a market's value is increasing, firms will see this opportunity, therefore joining this market. In that case, the job market in this industry will develop better.

b. Industrial Changes

Aside from government policy changes, the changes among firms in the industry may bring huge changes in career trends too. For example, when Ford invented the assembly line, the industry of cars changed amazingly fast, leading to prosperity in the industry of cars, and the improved way of producing brought millions of car assembling jobs[30]. Ai technology is improving at a fast speed, and it very will likely bring new changes like these to a lot of industries. Therefore, industry changes are considered a key part of industry transformations.

4.2 Database

We collected the following data online to enhance our paper. Data are collected for the three primary indicators and respective secondary indicators.

Our data focuses on two main aspects: **popularity** and **job vacancies**. Popularity is measured by weekly and monthly search volumes, Baidu views, and the number of related web pages, which show how often these careers are searched and discussed online. Job vacancies are assessed using the CIER index, indicating the balance between job openings and job seekers. Together, these indicators provide a clear view of market interest

and employment opportunities for each emerging profession.

Our data focuses on four key areas of industry potential. Industrial Growth is shown by the **Average Market Size Growth Rate**, indicating how fast each profession's market is expanding. **Talent Development** is measured by the Graduate Employment Tendency, showing the percentage of graduates choosing these careers. Technological Advancement is represented by **R&D Investment**, reflecting the amount spent on research and development in each field. Fungibility is captured by the **Replaceability Index**, showing how easily a job can be replaced. These indicators give a clear snapshot of growth, talent attraction, innovation, and job security for each profession.

The data shows the **proportion of government PPP funding (%)** for each career. Higher percentages, like for Intelligent Connected Vehicle Tester (0.318 %) and Non-Ferrous Metal Spot Trader (0.289%), reflect strong government support, while lower ones, like Energy Storage Station Maintenance Administrator (0.005%), indicate minimal backing. This highlights the level of public-private funding directed toward each profession's industrial transformation.

4.3 Normalization

First, we applied a **logarithmic transformation** to several key indicators to handle the large differences in magnitude. Specifically, we took the logarithmic values of the five **popularity indicators**—**Weekly Search Volume**, **Monthly Search Volume**, **Baidu Views**, **Number of Related Web Pages**, and the **CIER Index**—as well as the **R&D Investment under Technological Advancement**. This step helped to linearize the data and reduce the scale differences, making it more manageable for comparison and further analysis.

Next, we used **Proportional Transformation Method** to scale the remaining indicators (except for the **Replaceability Index**) down to a comparable range between 0 and 1. Min-Max Normalization is defined by the formula:

$$a_{ij}^* = \frac{a_{ij}}{\max_{1 \leq i \leq n} a_{ij}} \left(\max_{1 \leq i \leq n} a_{ij} \neq 0, 1 \leq i \leq n, 1 \leq j \leq m \right)$$

Finally, since the Replaceability Index is a **negative indicator** (i.e., lower values are better), we apply the following formula:

$$a_{ij}^* = \frac{\min a_{ij}}{a_{ij}} (1 \leq i \leq n, 1 \leq j \leq m)$$

This reverses the index, which ensures that lower original values are scaled higher, thus aligning it with other positive indicators.

4.4 Indicator Weights and AHP

4.4.1 Methodology

The **Analytic Hierarchy Process (AHP)**, introduced by operational researcher Thomas Saaty in the 1970s, is a decision-making method that combines qualitative and quantitative analysis. It is particularly useful for complex problems with multiple criteria, especially when vast datasets for objective weighting are unavailable.[31]. We use AHP because it allows us to quantify subjective judgments, making it suitable when precise measurement of relationships is difficult due to a lack of extensive data. AHP helps simplify decision-making by structuring and prioritizing various indicators of a problem.

In AHP, we first determine the primary and secondary indicators (which we already finished). We will conduct AHP on indicators of the same level. Next, we construct a **pairwise comparison matrix**, where each element is compared with others using a scale from 1 to 9, indicating relative importance. The comparison matrix A is used to calculate the eigenvector corresponding to the largest eigenvalue $\lambda_{\max}$, which represents the relative weights of the criteria. Finally, we calculate the **weights** by normalizing the eigenvector. The weight vector W is derived from the equation $AW = \lambda_{\max} W$, giving the weight of each criterion. Finally, to verify the consistency of the comparisons, we perform a Consistency Check using the Consistency Ratio (CR). The formula is:

$$CR = \frac{CI}{RI}$$

where $CI = \frac{\lambda_{\max} - n}{n - 1}$ is the Consistency Index. A CR value less than 0.1 indicates acceptable consistency in the comparisons.

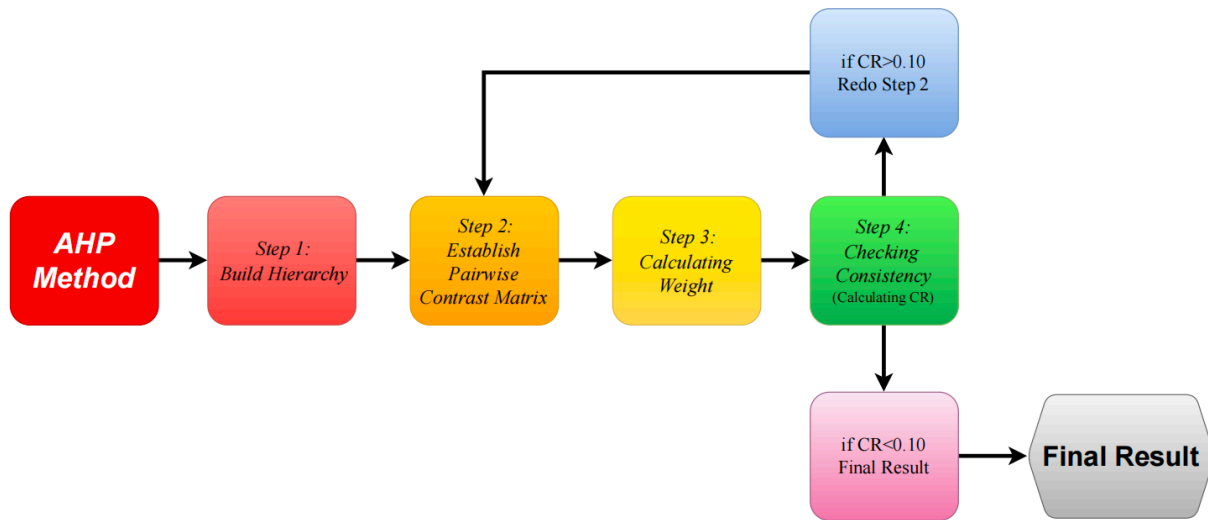


Figure 4.1: AHP Method

4.4.2 AHP Process and Results

We need 3 AHP models to handle the different layers of our data. Since we have 4 primary indicators and specific data sets—5 indicators for Market Supply-Demand Balance and 4 for Industry Potential—we construct separate AHP models for each layer. The first AHP model compares the primary indicators, while the second focuses on the Market Supply-Demand Balance indicators and the third on the Industry Potential indicators.

According to our understanding of what is more illustrative of Development Trends, we will determine the relative importance of the four primary indicators. The following are our tables:

	Supply-Demand	Ind. Potential	Ind. Trans.
Supply-Demand	1	$\frac{1}{2}$	2
Ind. Potential	2	1	4
Ind. Trans.	$\frac{1}{2}$	$\frac{1}{4}$	1

The first is the relationships between primary indicators. It is revealed in the table that we deem **Industrial Potential** to be the most important among the three. This is not merely our opinion. In fact, according to the World Economic Forum (2023), technological advancements and industrial growth, especially in sectors like AI, renewable energy, and digital trade, are expected to drive significant job creation over the next five years.[32]

After calculation, the results are demonstrated in the following table. The values for the Consistency Index

Indicator	Value
Consistency Index (CI)	$-4.440892098500626e^{-16}$
Consistency Ratio (CR)	$-7.656710514656253e^{-16}$

Table 4.1: Consistency Index and Consistency Ratio values of Primary Indicators

(CI) and Consistency Ratio (CR) are extremely close to zero, which means that the pairwise comparisons in the AHP model are highly consistent. According to the model, the weights of the primary indicators are as follows:

Next, we will repeat a similar process for the secondary indicators used to calculate Market Supply-Demand Balance. The pairwise comparison tables, CR value, and Weights are indicated as follows:

Finally, pairwise comparison tables and CR value of Industry Potential are indicated as follows:

Primary Indicator	Weight
Market Supply-Demand Balance	0.2857
Industry Potential	0.5714
Industrial Transformation	0.1429

Table 4.2: Weights for the Three Primary Indicators

	Wk. Search	Mo. Search	Baidu Views	Web Pages	CIER Index
Wk. Search	1	$\frac{1}{2}$	1	1	$\frac{1}{4}$
Mo. Search	2	1	1	1	$\frac{1}{3}$
Baidu Views	1	1	1	1	$\frac{1}{3}$
Web Pages	1	1	1	1	$\frac{1}{4}$
CIER Index	4	3	3	4	1

Secondary Indicator	Weight
Average Weekly Search Volume	0.1143
Average Monthly Search Volume	0.1596
Baidu Encyclopedia Views	0.1373
Number of Related Web Pages	0.1296
CIER Index	0.4592

Table 4.3: Weights for the Secondary Indicators

Metric	Value
Consistency Index (CI)	0.01389
Consistency Ratio (CR)	0.01239

Table 4.4: Consistency Index and Consistency Ratio for Secondary Indicators

	Size Growth	Employment	R&D Invest.	Replaceability
Size Growth	1	1	1/3	1/3
Employment	1	1	1/3	1/3
R&D Invest.	3	3	1	1/3
Replaceability	3	3	3	1

Metric	Value
Consistency Index (CI)	0.0515
Consistency Ratio (CR)	0.0572

Table 4.5: CI and CR for Secondary Indicators

4.5 TOPSIS

We decide to use **TOPSIS** to calculate the final results of career trends. A detailed description of its methodology is given in previous sections.

4.6 Results and Analysis

4.6.1 Career Trend Result

After calculations, the results are demonstrated as follows, with scores from the greatest to the least:

4.6.2 Analysis

Since there is a clear outlier in the score, we will use a boxplot to analyze the result of career trend scores. An analysis of the boxplot indicates that most career scores fall within the range of 20 to 65. This distribution is considered reasonable in the context of our application of the TOPSIS method. The methodology inherently

Career	Score
Non-Ferrous Metal Spot Trader	60.23
User Growth Operations Specialist	58.51
Intelligent Manufacturing System Maintenance Technician	52.26
Industrial Internet Maintenance Technician	50.71
Cybersecurity Grading and Assessment Specialist	46.48
Cloud Network Intelligent Maintenance Technician	46.12
Intelligent Connected Vehicle Tester	41.53
Intelligent Connected Vehicle Installation and Maintenance Technician	41.16
Bioengineering Technician	40.26
Live Streamer	39.17
Hydrogen-Based Direct Reduction Iron Worker	38.55
Generative AI System Application Specialist	37.15
Copyright Agent	36.39
Energy Storage Station Maintenance Administrator	31.63
Power Quality Administrator	30.53
Cultural and Creative Product Planning and Operations Specialist	28.84
Ski Patrol and Rescue Technician	28.22
Oral Health Technician	24.25
Exhibition Builder	8.08

Table 4.6: Career Trend Scores with (Top 5 Highlighted)

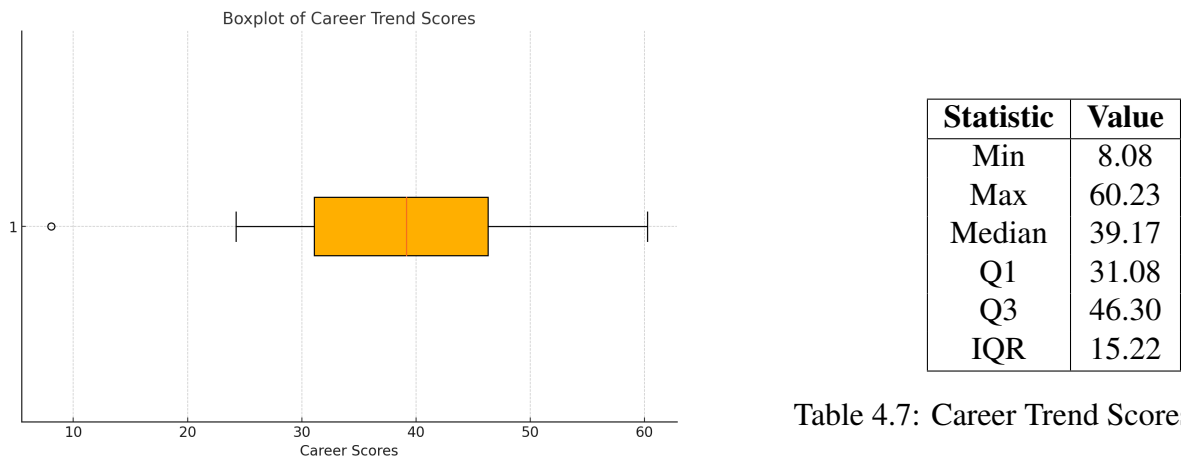


Table 4.7: Career Trend Scores Statistics

Figure 4.2: Career Trend Boxplot

acknowledges that most professions exhibit a balance of advantages and disadvantages. It is virtually impossible for a career to excel uniformly across all evaluated dimensions—namely popularity, growth potential, job vacancies, and industrial transformation—due to the multifaceted nature of occupational assessments.

The career that emerges at the top of our ranking is the **Non-Ferrous Metal Spot Trader**. This profession attains the highest score primarily because of its substantial industry potential and a low replaceability score, indicating a high demand for specialized skills that are not easily automated or outsourced. Following closely are several internet-related professions such as the **User Growth Operations Specialist** and the **Intelligent Manufacturing System Maintenance Technician**. This pattern reflects a broader trend in the current job market, where digitalization is increasingly gaining prominence and offers significant potential for future development. The ascendancy of these careers aligns with prevailing industry dynamics and societal shifts toward technology-driven solutions.

In contrast, the **Exhibition Builder** stands out as a notable outlier with an exceptionally low score of approximately 8. This markedly low rating can be attributed to the limited market size and an unfavorable supply-demand balance within this sector. The scarcity of opportunities and minimal growth prospects in this

field result in a lower overall evaluation compared to other emerging careers analyzed in our study.

4.7 Sensitivity Analysis

We will test the results' sensitivity to the variables that we held constant. Specifically, we alter the original values of the indicators to see how the results will change. The following table demonstrates the sensibility of our result in respect to changes in secondary indicators. (+/-10%) Positive sensitivity values indicate that an increase in the indicator results in an increase in the output, while negative values indicate a decrease in the output when the indicator increases.

Secondary Indicator	Change in Result
Weekly Search Volume	4.42×10^{-5}
Monthly Search Volume	4.80×10^{-5}
Popularity (Baidu Views)	-7.78×10^{-6}
Number of Related Web Pages	8.10×10^{-5}
CIER Index	8.10×10^{-18}
Industrial Growth Rate	0.000
Graduate Employment Tendency	0.000
Technological Advancement (Patent Development)	1.63×10^{-3}

Table 4.8: Sensitivity Analysis

5 Strength and Weakness

Strength: The strength of our model lies in its use of diverse indicators to capture multiple facets of career development trends. The Coefficient of Variation (CV) method enhances differentiation by magnifying variability, ensuring significant differences are visible and accurately assessed. Combined with TOPSIS, this approach highlights disparities in secondary indicators, providing a nuanced analysis that aligns with industry trends and offers valuable insights for graduates and policymakers.

However, there are also some potential weaknesses. Though our data encompasses several aspects of emerging careers, it is still not enough. In real life, many other factors can affect the real result, such as the economic development of country.

6 Providing Career Advice for Graduates

After analyzing the long-term career development trends, it's essential to provide practical and personalized advice for graduates. To achieve this, we developed a model designed to address the following critical questions:

1. How does a graduate's background influence career path choices?

This considers factors such as educational qualifications, skill sets, and personal interests to identify how different backgrounds align with emerging careers.

2. Which emerging careers offer the best long-term potential for a graduate given their specific background?

Based on individual profiles and our analysis of career development trends, we suggest careers that combine the best market opportunities with the graduate's unique strengths and interests.

6.1 Individual Survey and Example

We recognize that individual backgrounds vary among graduates. That is, it becomes illogical to give personalized results without a graduate's specific interests and background. Therefore, an individual survey is designed to trace the interest and background. Our survey is illustrated below. The survey asks the graduate student to select the top 5 choices within the 19 emerging professions, and then select the expected salary that they are to find.

6.2 Matching the Students With The Careers

It is then the turn of the profession to either **accept or refuse** the graduate student. If the student acquires skills entailed in the career and provides the appropriate salary that graduate students expect, we say that the student and the career **match**", meaning the graduate student likes the career, and the career is 'willing to' accept the student.

Figure 6.2 provides an example when a student completes the individual survey. First, the graduate provides personal information by using the table above. The information he gives looks like the graph below, *Final results*.

After running the code, the result is directly provided with a single career. In the example, the output is called: Cybersecurity Grading and Assessment Specialist.

Career(s)	Order	Career	Order	Salary Expectation	Selection
Intelligent Connected Vehicle Tester		Non-Ferrous Metal Spot Trader		2500 ~ 5000	
Intelligent Connected Vehicle Installation and Maintenance Technician		Exhibition Builder		5001 ~ 7500	
Cloud Network Intelligent Maintenance Technician		Cultural and Creative Product Planning and Operations Specialist		7501 ~ 10000	
User Growth Operations Specialist		Energy Storage Station Maintenance Administrator		10001 ~ 12500	
Live Streamer		Power Quality Administrator		12501 ~ 15000	
Bioengineering Technician		Copyright Agent		15001 ~ 17500	
Oral Health Technician		Ski Patrol and Rescue Technician		17501 ~ 20000	
Cybersecurity Grading and Assessment Specialist		Hydrogen-Based Direct Reduction Iron Worker		20001 ~ 25000	
Generative AI System Application Specialist		Intelligent Manufacturing System Maintenance Technician		25000+	
Industrial Internet Maintenance Technician		Please Select All of the Careers You Interested in.		/	Only 1 Selection

Figure 6.1: Personal Survey

Career(s)	Order	Career	Order	Salary Expectation	Selection
Intelligent Connected Vehicle Tester		Non-Ferrous Metal Spot Trader		2500 ~ 5000	
Intelligent Connected Vehicle Installation and Maintenance Technician		Exhibition Builder		5001 ~ 7500	
Cloud Network Intelligent Maintenance Technician	1	Cultural and Creative Product Planning and Operations Specialist		7501 ~ 10000	1
User Growth Operations Specialist		Energy Storage Station Maintenance Administrator		10001 ~ 12500	
Live Streamer		Power Quality Administrator		12501 ~ 15000	
Bioengineering Technician		Copyright Agent		15001 ~ 17500	
Oral Health Technician		Ski Patrol and Rescue Technician		17501 ~ 20000	
Cybersecurity Grading and Assessment Specialist	1	Hydrogen-Based Direct Reduction Iron Worker		20001 ~ 25000	
Generative AI System Application Specialist	1	Intelligent Manufacturing System Maintenance Technician	1	25000+	
Industrial Internet Maintenance Technician	1	Please Select All of the Careers You Interested in.		/	Only 1 Selection

Figure 6.2: Example of student

6.3 Strengths and Weaknesses

The strength of the model of selecting a profession is clear:

Firstly, the model prioritizes the background and interests of the graduate students, ensuring that the selected result fits the personal choice of the graduate student.

Secondly, the model effectively uses the results from previous career trends to make suggestions. This guarantees that the graduate student will not reach an embarrassed state where the career they choose, though aligning with their interests, fails to account for the long-term potential.

However, some weaknesses are evident in our model.

The algorithm is limited to certain types of students. Graduate students need to have at least some interest in the emerging careers. Say, for instance, a graduate student is more into the field of accountant, which is a common career not indicated in the emerging careers. In this circumstance, our model can only give suggestions regarding the best fit career in this 19 career pool, but not all industries in real circumstances.

7 Science Article

Emerging Careers in China: How New Professions Shape the Future Job Market

In recent years, China has introduced a wave of new professions that promise to shape the future of its workforce. In July 2024, 19 new careers were officially recognized, ranging from *biotechnologists* to *generative AI specialists*. These roles signal not only an evolving job market but also new opportunities for graduates seeking careers that align with the needs of a modern, digital economy.

However, choosing the right career can be a complex decision. Emerging professions, while exciting, come with uncertainties. What does the future hold for these careers? Which ones are likely to thrive, and how

will they impact society?

A Data-Driven Approach to Career Prospects

One way to understand these new professions is by examining *both their immediate and long-term potential*. Factors such as *market demand, income levels, and working conditions* play a major role in determining whether a career path is sustainable.

For example, roles like *Cloud Network Intelligent Maintenance Technician* and *Live Streamer* are gaining traction due to the increasing reliance on cloud technologies and digital platforms. Jobs in these fields are not only in demand but also offer competitive income and flexible working conditions. However, the success of a career isn't just about what happens today. Graduates must consider long-term prospects, such as *industry growth, innovation, and job security*.

What Makes a Career "Future-Proof"?

As we look ahead, careers that align with technological advancements and government policy support are likely to thrive. Professions like *User Growth Operations Specialist* and *Intelligent Manufacturing System Maintenance Technician* are excellent examples. These jobs are integral to industries that are transforming rapidly due to digitalization and automation.

Another crucial factor is *fungibility*, or the risk of being replaced by machines or artificial intelligence. Jobs that require human creativity, problem-solving, and technical expertise are less likely to be automated. This makes roles in sectors like cybersecurity and intelligent manufacturing particularly resilient.

The Top Five Emerging Careers for Long-Term Growth

Based on trends and data, the five careers expected to experience the most significant long-term growth in China are:

1. **Non-Ferrous Metal Spot Trader:** As the global demand for metals increases, this role is expected to become more critical for resource management.
2. **User Growth Operations Specialist:** As digital platforms expand, experts who can drive user engagement and growth will be in high demand.
3. **Intelligent Manufacturing System Maintenance Technician:** With more industries adopting automated systems, professionals in this field will play a key role in keeping operations running smoothly.
4. **Industrial Internet Maintenance Technician:** This role supports the infrastructure that enables industries to connect their machines and operations through the Internet.
5. **Cybersecurity Grading and Assessment Specialist:** As cyber threats continue to rise, experts in this field will be vital for ensuring the safety of digital systems.

How These Careers Impact Society

The rise of these new professions reflects a broader shift in China's economy toward technology and innovation. As industries like renewable energy, artificial intelligence, and advanced manufacturing grow, they will require a highly skilled workforce. Graduates entering these fields will not only enjoy career stability but will also contribute to the country's economic progress and technological leadership.

These emerging professions are more than just jobs—they represent the future of work in China. By aligning education, skills, and interests with market demands, graduates can secure rewarding careers that offer both personal and societal benefits.

In a rapidly changing job market, understanding which professions are on the rise—and why—is crucial. As we look ahead, it's clear that careers driven by technology and innovation will shape the future of work, providing exciting opportunities for those ready to embrace them.

8 Conclusion

Based on the scores, we can generalize that careers with strong ties to technology-driven industries and digital transformation are more likely to have a favorable long-term development trend. For example, roles such

as Non-Ferrous Metal Spot Trader, User Growth Operations Specialist, and Intelligent Manufacturing System Maintenance Technician show higher scores indicating their importance in sectors that are either industrially vital or tied to digital growth and emerging technologies. These industries are likely to expand due to increasing global demand for technology integration, data-driven operations, and efficient resource management.

On the other hand, careers with lower scores, such as Exhibition Builder and Oral Health Technician, might experience more stagnant growth due to their limited scope in comparison to rapidly growing tech fields. These roles, while still valuable, might not benefit as much transformation, digital transformation, or advancements in AI as others.

However, there are still some limitations on the models. For instance, the model's overall computational approach appears overly simplistic, with an evident lack of sufficient data to support robust analysis and decision-making. Due to the various situations in reality, the assumptions made before modeling simplified the complicated questions, which leads to greater gaps to actuality.

References

- [1] Yujie, M. X. and C. (2023, August 18). A deep dive into labour demand in China - china business knowledge. China Business Knowledge -. <https://cbk.bschoo1.cuhk.edu.hk/a-deep-dive-into-labour-demand-in-china>
- [2] Microeconomics, 9th edition. Microeconomics. (n.d.). <https://www.pearson.com/us/higher-education/product/Pindyck-Microeconomics-9th-Edition/9780134184241.html>
- [3] Kim, B.-J. (2020). Unstable Jobs Harm Performance: The Importance of Psychological Safety and Organizational Commitment in Employees. *SAGE Open*, 10(2), 215824402092061. <https://doi.org/10.1177/2158244020920617>
- [4] Nghia, T. L. H., Bui, B. C., Singh, J. K. N., Lu, V. N. (1970, January 1). From employability to employment outcomes and Career Advancement: A Literature Review. SpringerLink. <https://link.springer.com/chapter/10.1007/978-981-19-3959-42>
- [5] Bronkhorst, B. (2015). Behaving safely under pressure: The effects of job demands, resources, and safety climate on employee physical and psychosocial safety behavior. **Journal of Safety Research**, 55, 63-72. <https://doi.org/10.1016/j.jsr.2015.09.007>
- [6] Thomas, S. (2023, April 12). The 5 determinants of demand explained: Outlier. Outlier Articles. <https://articles.outlier.org/determinants-of-demand>
- [7] Job demands - OSHwiki | European Agency for Safety and Health at Work. (2013, May 30). Europa.eu. <https://oshwiki.osha.europa.eu/en/themes/job-demands>
- [8] Golden, L. (2012). The Effects of Working Time on Productivity and Firm Performance, Research Synthesis Paper.
- [9] Blanchard, O., Diamond, P. (1989). The Beveridge Curve. **Brookings Papers on Economic Activity**, 1989(1), 1-60.
- [10] Celes. (2016, April 12). The Law of Diminishing Returns - Personal Excellence. Personalexcellence.co. <https://personalexcellence.co/blog/diminishing-returns/>
- [11] Burke, R. J., Singh, P., & Fiksenbaum, L. (2010). Work intensity: potential antecedents and consequences. *Personnel Review*, 39(3), 347-360.
- [12] McHill, A. W., & Wright Jr, K. P. (2019). Cognitive impairments during the transition to working at night and on subsequent night shifts. *Journal of biological rhythms*, 34(4), 432-446.
- [13] Cai, M., Wang, W., Cui, Y., & Stanley, H. E. (2018). Multiplex network analysis of employee performance and employee social relationships. *Physica A: Statistical Mechanics and Its Applications*, 490, 1-12.
- [14] Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. University of Chicago Press.
- [15] Mortensen, D. T., & Pissarides, C. A. (1998). Technological progress, job creation, and job destruction. *Review of Economic dynamics*, 1(4), 733-753.
- [16] Balsmeier, B., & Woerter, M. (2019). Is this time different? How digitalization influences job creation and destruction. *Research policy*, 48(8), 103765.
- [17] Rastgoo, P. (2016). The relationship of talent management and organizational development with job motivation of employees. *Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis*, 64(2), 653-662.
- [18] Market Potential. (2024). Baidu Encyclopedia. <https://baike.baidu.com/item/>

- [19] Su Yanzhao Wang Guanghui. (n.d.). Retrieved September 29, 2024 <http://iple.cass.cn/ldjjyj/1w/202201>
- [20] National Data. (2012). Stats.gov.cn <https://data.stats.gov.cn/easyquery.htm?cn=A01>
- [21] Government of Canada, S. C. (2018, December 10). Assessing Job Quality in Canada: A Multidimensional Approach. Ww150.Statcan.gc.ca. <https://www150.statcan.gc.ca/n1/pub/11f0019m/11f0019m2018412-eng.htm>
- [22] Macroeconomics. (2024). Baidu Encyclopedia. <https://baike.baidu.com/item>
- [23] Aaronson, D., & Phelan, B. J. (2019). Wage shocks and the technological substitution of low-wage jobs. *The Economic Journal*, 129(617), 1-34.
- [24] The Industrial Revolution (2020). Baidu Baike <https://baike.baidu.com/item/>
- [25] Notice of the General Office of the State Council on Issuing the Development Plan for the New Energy Vehicle Industry (2021-2035) (n.d.). https://www.gov.cn/zhengce/content/2020-11/02/content_5556716.htm
- [26] Barro, R. J. (1997). *Macroeconomics*. MIT Press.
- [27] Lee, S., Schmidt-Klau, D., Weiss, J., & Chacaltana, J. (2020). Does economic growth deliver jobs? Revisiting Okun's Law (No. 17). ILO Working Paper.
- [28] Radermacher, A., Walia, G., & Knudson, D. (2014, May). Investigating the skill gap between graduating students and industry expectations. In *Companion Proceedings of the 36th international conference on software engineering* (pp. 291-300).
- [29] Spitzer, B., Morel, V., Buvat, J., & Kanakadandi, S. (2015). THE DIGITAL TALENT GAP. DEVELOPING SKILLS FOR TODAY'S DIGITAL ORGANIZATIONS. In *ICERI2015 Proceedings* (pp. 3488-3499). IATED.
- [30] Henry Ford (2024). Baidu Baike <https://baike.baidu.com/item/>
- [31] Mathematical models Model Evaluation Model 08 Analytic Hierarchy Process (n.d.) <https://reformship.github.io/pages/1capacity/1model>
- [32] The Future of Jobs Report 2023. World Economic Forum. (n.d.). <https://www.weforum.org/publications/the-future-of-jobs-report-2023/digest>
- [33] The ideal solution of Topsis Method | Download scientific diagram. (n.d.). <https://www.researchgate.net/figure>

A Appendix

Career	Industry Potential			
	Industrial Growth	Talent Development	Technological Advancement	Fungibility
	Average Market Size Growth Rate (%)	Graduate Employment Tendency (%)	R&D Investment (100 million CNY)	Replaceability Index
Intelligent Connected Vehicle Tester	0.47	17.3	105.32	0.59
Intelligent Connected Vehicle Installation and Maintenance Technician	0.47	17.3	105.32	0.59
Cloud Network Intelligent Maintenance Technician	0.2	56.4	54.7	0.21
User Growth Operations Specialist	0.11	16.4	20	0.13
Live Streamer	0.55	48.2	20	0.39
Bioengineering Technician	0.14	19.1	829.2	0.3
Oral Health Technician	0.18	19.1	43	0.51
Cybersecurity Grading and Assessment Specialist	0.11	56.4	67.8	0.21
Generative AI System Application Specialist	0.84	45.5	135	0.36
Industrial Internet Maintenance Technician	0.04	56.4	581.34	0.21
Non-Ferrous Metal Spot Trader	0.08	16.4	800	0.17
Exhibition Builder	0.18	5.5	4.1	0.63
Cultural and Creative Product Planning and Operations Specialist	0.13	16.4	7.96	0.25
Energy Storage Station Maintenance Administrator	0.3	56.4	120	0.85
Power Quality Administrator	0.05	5.5	1082	0.85
Copyright Agent	0.05	5.5	8.48	0.25
Ski Patrol and Rescue Technician	0.06	5.5	3.05	0.67
Hydrogen-Based Direct Reduction Iron Worker	0.09	5.5	500	0.83
Intelligent Manufacturing System Maintenance Technician	0.15	56.4	843	0.21

Figure A.1: Data for Industry Potential

Career	Market Supply-Demand Balance				
	Popularity				Job Vacancy
	Weekly Search Volume	Monthly Search Volume	Baidu Views (times)	Number of Related Web Pages (thousands)	CIER Index
Intelligent Connected Vehicle Tester	91	410	1249	1630	1.87
Intelligent Connected Vehicle Installation and Maintenance Technician	28	122	461	135	1.83
Cloud Network Intelligent Maintenance Technician	28	134	988	869	3.67
User Growth Operations Specialist	133	575	1755	455	5.71
Live Streamer	420	1828	727820	10300	5.71
Bioengineering Technician	63	279	1418	3950	2.3
Oral Health Technician	868	3746	9748	378	2.3
Cybersecurity Grading and Assessment Specialist	98	431	1647	797	3.67
Generative AI System Application Specialist	98	426	2430	1130	2.48
Industrial Internet Maintenance Technician	35	159	717	1630	3.67
Non-Ferrous Metal Spot Trader	21	108	1051	451	0.47
Exhibition Builder	42	201	632	266	0.85
Cultural and Creative Product Planning and Operations Specialist	140	625	1859	372	0.85
Energy Storage Station Maintenance Administrator	35	153	607	267	3.67
Power Quality Administrator	21	115	662	496	0.47
Copyright Agent	70	313	12909	750	4.45
Ski Patrol and Rescue Technician	28	136	233	145	5.71
Hydrogen-Based Direct Reduction Iron Worker	14	66	383	27.9	0.47
Intelligent Manufacturing System Maintenance Technician	42	204	792	1290	3.91

Figure A.2: Data for Market Supply-Demand Balance

Career	Industrial Transformation
	Policy
	Proportion of Government PPP Funding (%)
Intelligent Connected Vehicle Tester	0.318
Intelligent Connected Vehicle Installation and Maintenance Technician	0.318
Cloud Network Intelligent Maintenance Technician	0.006
User Growth Operations Specialist	0.006
Live Streamer	0.013
Bioengineering Technician	0.006
Oral Health Technician	0.014
Cybersecurity Grading and Assessment Specialist	0.006
Generative AI System Application Specialist	0.006
Industrial Internet Maintenance Technician	0.006
Non-Ferrous Metal Spot Trader	0.289
Exhibition Builder	0.032
Cultural and Creative Product Planning and Operations Specialist	0.013
Energy Storage Station Maintenance Administrator	0.005
Power Quality Administrator	0.005
Copyright Agent	0.006
Ski Patrol and Rescue Technician	0.007
Hydrogen-Based Direct Reduction Iron Worker	0.289
Intelligent Manufacturing System Maintenance Technician	0.006

Figure A.3: Data for Industrial Transformation